

FinPulse: Master Learning Guide

Introduction: Building Modern Apps

To build a project like FinPulse, you need a mix of Backend, Frontend, and Database skills. This guide breaks down exactly what you need to master.

Phase 1: Backend Mastery (FastAPI)

1. REST API Basics: Understanding GET, POST, PUT, DELETE requests.
2. FastAPI Framework: Learning how to create routes, handle JSON data, and use Pydantic for validation.
3. Security (JWT): JSON Web Tokens are used for login sessions. You need to learn how to create and verify tokens.
4. Bcrypt: Always hash passwords! Never store plain text passwords in your database.
5. Dependency Injection: Using 'Depends' in FastAPI to manage database connections and user authentication.

Phase 2: Database (PostgreSQL & SQL)

1. Relational Design: Creating tables (Accounts, Users, Expenses) and linking them with IDs.
2. SQL Queries: Mastering SELECT (fetching), INSERT (adding), and JOIN (linking tables) queries.
3. Timezone Management: Storing dates as UTC and converting them to local time (like UTC+5 for Uzbekistan).
4. Database Drivers: Using 'psycopg2' to let Python talk to the PostgreSQL database.

Phase 3: Frontend (JavaScript & Modern UI)

1. DOM Manipulation: Using JavaScript to update the screen without refreshing the page.
2. Fetch API: Communicating with your Backend to send/receive data.
3. CSS Mastery: Flexbox and Grid for layout, and Glassmorphism (blur/transparency) for that premium feel.
4. SPA (Single Page Application): Creating one HTML file and using JS to swap between 'Home', 'Expenses', and 'Admin' sections.

Phase 4: Integrations & Visualization

1. Email APIs: Using Brevo (or SendGrid/Mailchimp) to send automated verification codes.
2. Chart.js: A powerful library to create Pie, Bar, and Line charts using your financial data.
3. CSV Generation: Converting your data arrays into CSV files for Excel users.

FinPulse: Master Learning Guide

Phase 5: Pro Developer Tips

1. Environment Variables: Always use .env files for secrets (API keys, passwords).
2. Polling: Use setInterval() in JS to keep your dashboard updated in real-time.
3. Error Handling: Always try/except on the backend and catch errors on the frontend to keep the app from crashing.

FinPulse: Master Learning Guide

O'quv qo'llanma (Qisqacha o'zbekcha)

Ushbu loyihani noldan qurish uchun siz quyidagilarni o'rganishingiz kerak:

1. Python & FastAPI: API yaratish va JWT orqali xavfsizlikni ta'minlash.
2. SQL & PostgreSQL: Ma'lumotlarni jadvallarda saqlash va boshqarish.
3. JavaScript (ES6+): Sahifani yangilamasdan ma'lumotlar bilan ishlash (Fetch API).
4. CSS Dizayn: Zamonaviy 'Glassmorphism' va 'Dark Mode' dizaynlarini yaratish.
5. API Integratsiyalari: Brevo (Email yuborish) va Chart.js (Grafiklar chizish).
6. Xavfsizlik: Parollarni shifrlash (Bcrypt) va Admin paneli uchun maxfiy kirish tizimi.